

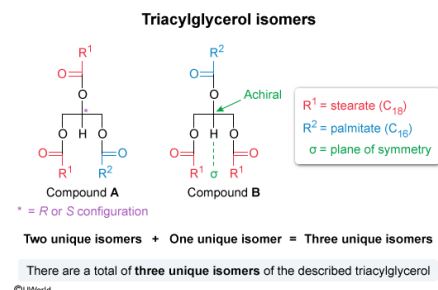
How many unique isomers are possible for a triacylglycerol molecule prepared from two molecules of stearate and one molecule of palmitate?

- ☐ A. 1 [6%]
☐ B. 2 [42%]
☒ C. 3 [37%]
☐ D. 4 [13%]

Correct

38% answered correctly

Explanation:



A **triacylglycerol** is a lipid prepared from three **fatty acyl** groups and a molecule of **glycerol**. Each fatty acyl group is bonded to the glycerol through an **ester** linkage. A single triacylglycerol molecule can contain up to three different fatty acyl groups; the composition and relative positions of acyl groups impacts the properties of the triacylglycerol.

Isomers are molecules with an identical molecular formula but with different connectivity or positioning of atoms. **Constitutional (structural) isomers** differ in the connectivity of atoms whereas **stereoisomers** differ in the way atoms are positioned in three-dimensional space.

This question asks how many unique isomers are possible for a triacylglycerol molecule prepared from two molecules of stearate (C_{18} fatty acid) and one molecule of palmitate (C_{16} fatty acid). Since glycerol is symmetrical, having the palmitate attached at *either end* of a glycerol results in the same constitutional isomer (Compound **A**). However, this bonding pattern generates a **stereocenter** at the central glycerol carbon. As a result, both the **R** and **S isomers** of Compound **A** are possible outcomes.

Having the palmitate attached to the middle carbon atom results in a third possible triacylglycerol isomer (Compound **B**). Since glycerol is symmetric, Compound **B** is **achiral**.

Therefore, a total of **three unique isomers are possible** through this preparation.

(Choice A) One unique triacylglycerol isomer can only happen when all three acyl groups are *identical*. Since there are two different acyl groups, this is no longer possible.

(Choice B) There would be two unique triacylglycerol molecules if only *constitutional isomers* are considered and *stereoisomers* are not considered.

(Choice D) One way to arrive at a solution of four unique isomers is if the stereochemistry of the middle carbon in Compound **B** is incorrectly included. Since this carbon has two identical groups, *R* or *S* isomers of Compound **B** are not possible.

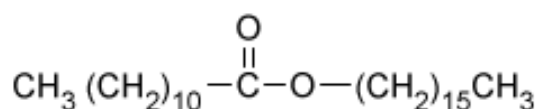
Educational objective:

Triacylglycerols are lipids that contain up to three different fatty acyl groups esterified to a glycerol. Triacylglycerol isomers can include both constitutional isomers and stereoisomers, depending on the fatty acyl group composition in the molecule.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 403917
System:
Functional Groups and Their Reactions

Which statement accurately describes a synthetic method to prepare the following wax?

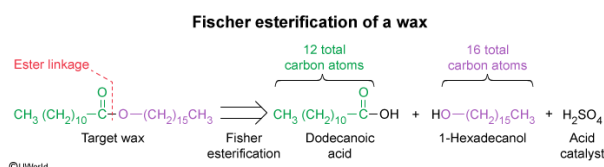


- ☐ A. Heat decanoic acid, 1-pentadecanol, and catalytic H_2SO_4 [16%]
- ☐ B. Heat 1-dodecanol, 1-hexadecanoic acid, and catalytic H_2SO_4 [14%]
- ☐ C. Heat dodecanoic acid and 1-hexadecanol [6%]
- ✓ ☒ D. Heat dodecanoic acid, 1-hexadecanol, and catalytic H_2SO_4 [62%]

Correct

63% answered correctly

Explanation:



Waxes are hydrophobic lipids structurally composed of a long-chain carboxylic acid (**fatty acid**) condensed with a long-chain **alcohol** through an **ester** linkage. An **esterification reaction** results in the formation of a new ester linkage. Direct esterification reactions between a carboxylic acid and an alcohol occur slowly at room temperature and in the absence of a **catalyst**. The **Fischer esterification** requires an acid catalyst (usually sulfuric acid, H_2SO_4) and heat, and is a **condensation reaction** that eliminates a molecule of water from the reactants alongside ester formation.

This question asks which statement correctly describes a method to prepare the depicted wax. The half of the wax containing the carbonyl ($\text{C}=\text{O}$) must be derived from a reactant carboxylic acid. Including carbons in the terminal methyl and carbonyl, a total of **twelve carbons** are in this half of the wax. The IUPAC prefix for twelve is *dodeca*–, and the reactant carboxylic acid is **dodecanoic acid**.

The other half of the wax is derived from a reactant alcohol. A total of **sixteen carbons** are in this half of the wax, arranged in a linear chain with the oxygen attached at carbon 1. The IUPAC prefix for sixteen is *hexadec*–; thus, the reactant alcohol is 1-hexadecanol. For an effective Fischer esterification, an acid catalyst (H_2SO_4) is required. Choice D lists all the necessary components to prepare the given wax molecule through a Fischer esterification.

(Choice A) Decanoic acid and 1-pentadecanol have prefixes that indicate molecules with a total of ten and fifteen carbon atoms, respectively. These numbers incorrectly use *only* the subscript values for the repeating methylene unit (CH_2), and do not include carbons from the terminal methyl groups and carbonyl.

(Choice B) The ester carbonyl is connected to the left alkyl chain, and is derived from the reactant carboxylic acid. This choice incorrectly exchanges the number of carbons from the reactant carboxylic acid and alcohol components.

(Choice C) This choice does not include the required acid catalyst (H_2SO_4) for a Fischer esterification.

Educational objective:

A wax can be prepared from carboxylic acid and alcohol reactants using an esterification reaction. A Fischer esterification requires heat and an acid catalyst to generate the new ester linkage.

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Subject:

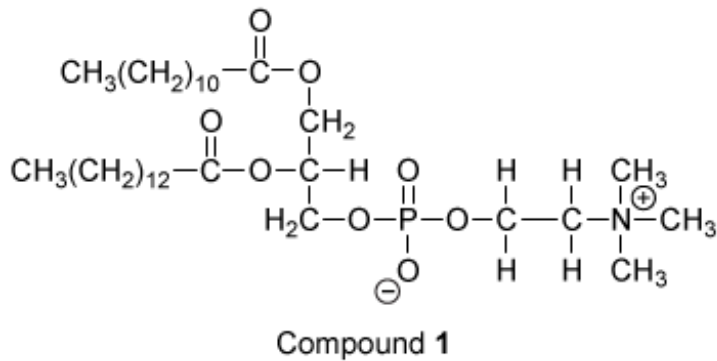
Organic Chemistry

QID:403918

System:

Functional Groups and Their Reactions

Compound **1** is a common component of plasma membranes.



How many unique types of soap molecules will be generated through the complete hydrolysis of Compound **1** using excess NaOH?

- ☐ A. 1 [4%]
- ✓ ☒ B. 2 [55%]
- ☐ C. 3 [35%]
- ☐ D. 5 [4%]

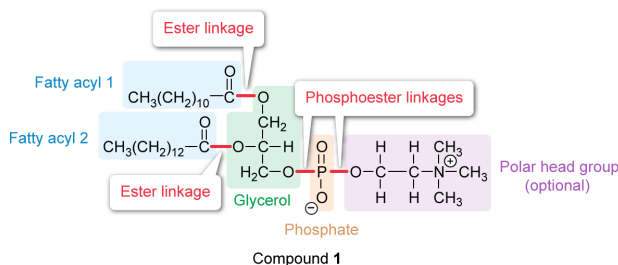
Correct

55% answered correctly

Explanation:

Generating soaps from phospholipids

General scaffold of a phospholipid



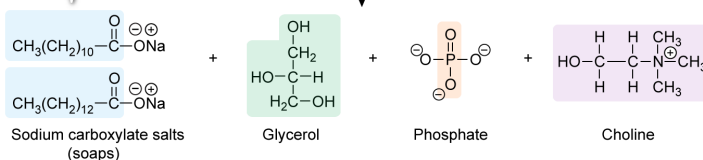
Different alkyl chains
= different soap molecules

Compound **1**

↓

Excess NaOH

Hydrolyzes all esters



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Soaps are defined as the sodium carboxylate salts of **fatty acids**.

Soaps are generally prepared by treating **fatty acyl**—containing lipid

sources (eg, triacylglycerols, phospholipids) with an excess of sodium hydroxide, which acts as a strong nucleophile to **hydrolyze** all **ester** linkages.

Phospholipids are a modular type of lipid, and are a major component of lipid bilayers. A simple **phospholipid** contains two fatty acyl groups esterified to alcohols on carbons 1 and 2 of glycerol, and an inorganic phosphate (PO_4^{3-}) esterified to the alcohol on carbon 3. The polar phosphate-containing region may be further esterified with other polar or charged small molecules to generate more complex phospholipid variants. The fatty acyl groups can be identical or different, depending on the identity of the acyl sidechains.

The structural features of Compound **1** indicates that it is a phospholipid variant, as it contains:

- Two fatty acyl groups
- A glycerol core
- An inorganic phosphate
- A small, charged organic group (**choline**)
- A total of four esters and phosphoesters linking the components

Treatment with excess NaOH will hydrolyze all ester and phosphoesters in Compound **1**, as well as deprotonate any acidic protons. Since the two fatty acyl groups contain a different number of carbon atoms, basic hydrolysis will lead to the formation of **two different sodium carboxylate salts (soaps)**.

(Choice A) One unique soap molecule would be produced if the fatty acyl groups in Compound **1** had *identical* structures. However, the fatty acyl groups in Compound **1** have different chain lengths.

(Choice C) Three soap molecules would be generated if Compound **1** was a triglyceride containing three *different* fatty acyl groups. However,

Compound **1** is a phospholipid.

(Choice D) Although the complete basic hydrolysis of Compound **1** generates five total molecular fragments, only *two* of these materials are classified as unique sodium carboxylate salts (soaps).

Educational objective:

A soap is a sodium carboxylate salt of a fatty acid. Soaps can be produced through basic hydrolysis of fatty acyl-containing lipids like triacylglycerols or phospholipids. The number of unique soap molecules that can be generated is dependent on the number of unique fatty acyl groups in the source lipid.

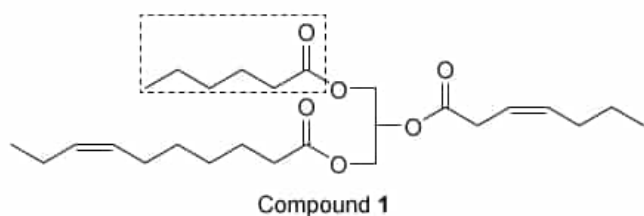
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Subject:
Organic Chemistry

QID:403919

System:
Functional Groups and Their Reactions

Which term most accurately describes the indicated region of Compound 1?

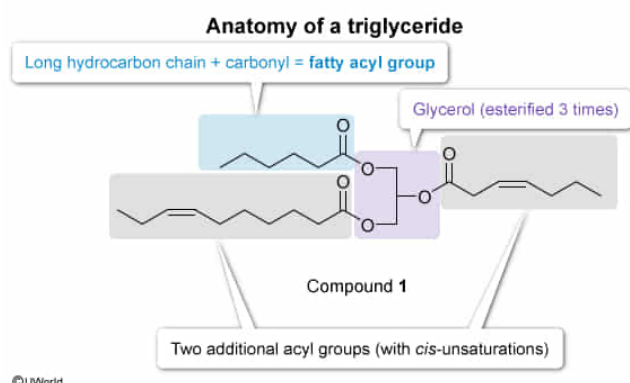


- ✔ ☒ A. Fatty acyl group [67%]
☐ B. Fatty acid [23%]
☐ C. Monoacylglycerol [7%]
☐ D. Triglyceride [1%]

Correct

67% answered correctly

Explanation:



Lipids are a structurally diverse class of biological molecules with shared traits of limited solubility in water and higher solubility in organic solvents. Lipid solubility profiles are primarily due to the presence of one or more structural regions composed of nonpolar hydrocarbons. Many types of lipids exhibit a **modular structure**, where discrete molecular components are connected through covalent bonds to generate larger lipid structures.

Fatty acids are among the most common simple lipids and consist of a long-chain **carboxylic acid** with zero to several *cis*-alkene functional groups within the hydrocarbon chain. The modular incorporation of a fatty acid into larger biomolecules generally occurs through either an **ester** or **amide** linkage generated through a **nucleophilic acyl substitution** reaction. A **fatty acyl group** refers to the carbonyl and hydrocarbon chain of a fatty acid that has been coupled to other biomolecules.

In this question, the indicated region includes the carbonyl of an ester and an additional hydrocarbon chain containing five carbon atoms. As such, this region would constitute an **acyl group** derived from a fatty acid containing a total of six carbon atoms (hexanoic acid). Consequently, the most appropriate term for this region is a **fatty acyl group**.

(Choice B) A fatty acid is a long-chain carboxylic acid or carboxylate. The indicated region is bonded to the rest of Compound **1** and does not include the complete carboxylic acid or carboxylate functional group.

(Choice C) A monoacylglycerol consists of a molecule of **glycerol** that has been acylated at only one of its three **hydroxyl groups**. The glycerol core in Compound **1** has been acylated at *three* hydroxyl groups, and the indicated region does not include the glycerol core.

(Choice D) The *totality* of Compound **1** is a **triglyceride**, whose definition exceeds the boundaries of the indicated region.

Educational objective:

Lipids are biomolecules that contain significant nonpolar characteristics. The correct use of lipid terminology is often related to the modular incorporation of lipid components within larger, more complex lipid variants.

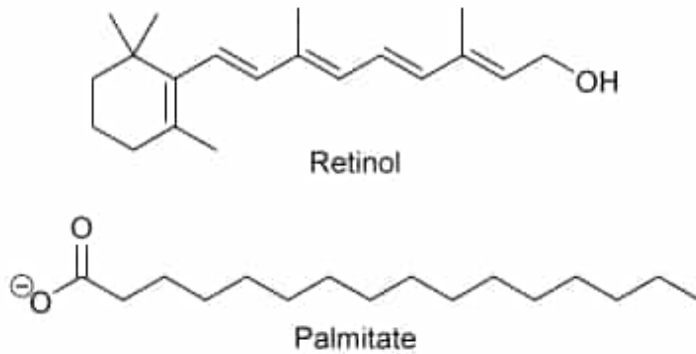
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Subject:
Organic Chemistry

QID:403920

System:
Functional Groups and Their Reactions

The lipids retinol and palmitate are shown below.



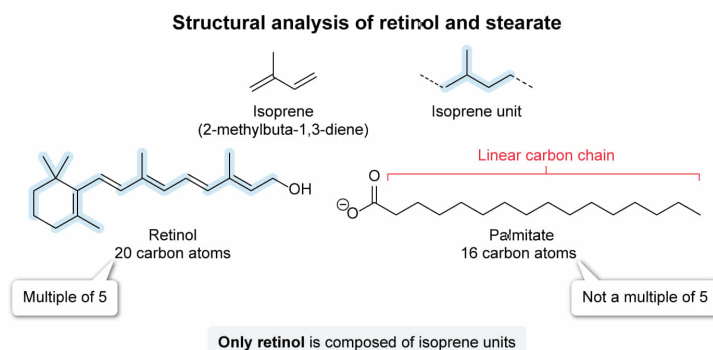
Which of these lipids is composed of isoprene units?

- ☐ A. Palmitate only [3%]
- ✓ ☒ B. Retinol only [92%]
- ☐ C. Both palmitate and retinol [2%]
- ☐ D. Neither palmitate nor retinol [2%]

Correct

91% answered correctly

Explanation:



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Terpenes are a broad class of lipids whose structures are comprised of two or more five-carbon groups called **isoprene units**. Isoprene units are named for their similarity to **isoprene**, which has a four-carbon linear chain with a one-carbon branch. During the biosynthesis of terpenes, individual isoprene units are joined in one of three ways: **head-to-head**, **tail-to-tail**, or **head-to-tail**. The presence and/or position of individual pi bonds may vary within terpenes.

Terpenes generally retain the following features:

- A total number of carbons that is a multiple of five
- A carbon skeleton that reflects the scaffold of an isoprene unit (unless a rearrangement such as a **methyl shift** has occurred)

This question asks which of the depicted lipids (retinol, palmitate) is structurally composed of isoprene units. Retinol has a total of 20 carbons, which is consistent with the requirement of maintaining a multiple of five carbon atoms. Retinol also has regular one-carbon branches within its structure, which supports the requirement of it being composed of isoprene units. The structure of retinol can also be mapped to depict locations of its four isoprene units.

By comparison, palmitate has a total of 16 carbon atoms, which is *not* a multiple of the five carbons in an isoprene unit. Furthermore, the carbon chain in palmitate is **linear** and lacks the necessary one-carbon branches characteristic of terpenes.

Consequently, **only retinol is composed of isoprene units.**

(Choice A) The structure of palmitate precludes its characterization as a terpene, as it does not contain an appropriate number of carbon atoms and contains a linear, unbranched carbon chain.

(Choice C) Although retinol meets the criteria of being composed of isoprene units, palmitate does not.

(Choice D) Although this solution correctly characterizes the relationship for palmitate, the number of carbons in retinol and its pattern of carbon chain branching *are* consistent with the composition of isoprene units.

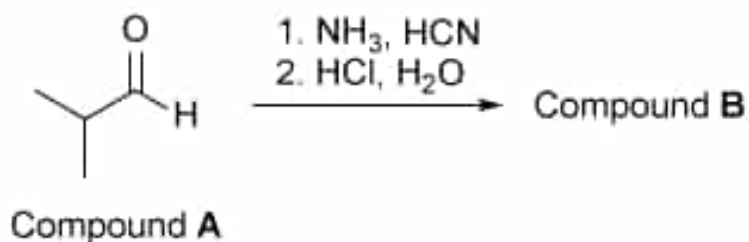
Educational objective:

Terpenes are natural lipid products derived from the five-carbon isoprene unit building block. Analysis of lipid structure provides clues about whether the lipid can be classified as a terpene.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403921
System:
Functional Groups and Their Reactions

Which statement most accurately describes the reaction sequence shown below?

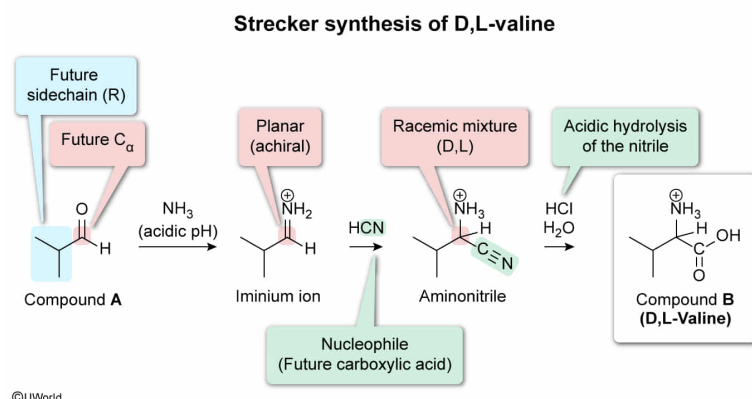


- ☐ A. Strecker synthesis of D,L-leucine [6%]
☐ B. Gabriel synthesis of L-valine [9%]
☐ C. Strecker synthesis of L-valine [40%]
☒ D. Strecker synthesis of D,L-valine [43%]

Correct

43% answered correctly

Explanation:



The molecular building blocks for peptides and proteins are **α -amino acids**. Each α -amino acid contains a carboxylic acid group, an α -amino group, and a **variable side chain (R)**. Most naturally occurring α -amino acids are biosynthesized as a single stereoisomer and have an **L-configuration**.

The **Strecker synthesis** is a man-made (synthetic) approach to prepare α -amino acids from aldehyde precursors. The carbon of

the aldehyde functional group ultimately becomes the α -carbon (C_α) in the final amino acid, and the aldehyde's alkyl chain becomes the side chain (R). Since the side chain in this question is an isopropyl group, the target α -amino acid is **valine**.

Treatment with ammonia (NH_3) generates a transient **iminium ion**, which then undergoes a **nucleophilic addition** reaction with a cyanide ion ($-CN$) to form an **aminonitrile**. The nucleophilic cyanide ion adds to either the top or bottom face of the planar (achiral) iminium ion and leads to a **racemic** outcome for the newly generated chiral carbon atom and a **D,L-amino acid product**.

Treatment of the aminonitrile intermediate with aqueous acid promotes hydrolysis of the nitrile functional group and its conversion to a carboxylic acid. Combining the features of this synthesis, this reaction sequence depicts the **Strecker synthesis of D,L-valine**.

(Choice A) The side chain for leucine is a four-carbon **isobutyl group**. In this question, the side chain precursor portion of Compound **A** is a three-carbon **isopropyl group**.

(Choice B) The **Gabriel synthesis** of α -amino acids utilizes a phthalimide-protected nitrogen component. These features are absent in the synthesis shown in this question.

(Choice C) In the Strecker synthesis, the planar iminium ion intermediate can be attacked by the cyanide ion from either face and will ultimately result in a *racemic* (D,L) amino acid product. This reaction cannot preferentially result in the formation of only the L-isomer.

Educational objective:

α -Amino acids can be prepared through the Strecker synthesis from appropriate aldehyde precursors. The products of a Strecker

synthesis will be racemic (D,L).

Time spent:0 sec

Subject:
Organic Chemistry

QID:403922

System:
Functional Groups and Their Reactions